Md. Anik Chowdhury

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Education

East West University **2022 – Present**
Bachelor of Science in Computer Science and Engineering; ***Cum. GPA: 3.46 / 4.0*** *Dhaka, Bangladesh*

Relevant Coursework

- OOP (JAVA)
- Data Structures
- Algorithms
- Software Engineering
- Software Testing and Quality Assurance
- Database Management (DBMS)
- Artificial Intelligence
- Machine Learning
- System Analysis and Design
- Computer Architecture
- Mobile Programming (Android)
- Web Programming

Projects

LenDen: A Seamless Mobile Banking Web Application| *PHP, MYSQL, React.js*

- Developed a full-featured mobile banking application providing secure account management, funds transfer, and bill payment functionalities.
- Implemented a robust PHP backend integrated with MySQL (via XAMPP) to handle efficient data management and secure transaction processing.
- Ensured data integrity and security by utilizing password encryption, PHP Sessions for user authentication, and encrypted communication protocols.
- Designed a responsive and user-friendly frontend using HTML, CSS, and JavaScript (React.js/Plain JS) for a convenient mobile banking experience.

Object Detection with SSL & Semi-Supervised Learning| *PyTorch, DINO, BYOL, YOLO, STAC*

- Pioneered an object detection system leveraging YOLOv10, YOLOv11, and YOLOv12 backbones to accurately detect fish fry for stocking density control and health assessment in aquaculture.
- Significantly enhanced model performance by integrating Self-Supervised Learning (SSL) methods, BYOL and DINO, for superior feature representation via pre-training on unlabeled data.
- Employed the STAC (Semi-Supervised Learning) framework to effectively utilize the limited labeled portion of the Fish Fry Dataset (FFD), boosting generalization and robustness.
- Benchmarked and structured training configurations across three YOLO versions and three learning strategies (BYOL, DINO, STAC), resulting in nine distinct experimental setups for comprehensive analysis.

Technical Skills

Languages: C, C++, Java, JavaScript, Typescript, HTML/CSS, SQL,Python **Developer Tools**: VS Code, Git and Github, Google Colab, Kaggle
Technologies/Frameworks: NodeJs, Linux, GitHub,

Awards and Achievements

Merit Scholarship **2022 – 2025**
East West University *Dhaka, Bangladesh*

- Maintained a CGPA of 3.50 or higher for multiple semesters.
- Tuition fee waiver for four academic year.

Research

Research Projects **2024 – Present**
East West University *Dhaka, Bangladesh*

- Explainable Machine-Learning Forecasts of Building-Energy Demand from Weather Signals Co-author. Published at IEEE International Conference on Smart Grid Communications (SmartGridComm) 2024 (Estimated based on current publication trend for this topic). DOI: 10.1109/SmartGridComm.2024.11172015.